INSPECTION ROBOT

Complete Technical Reference Manual

Mechanical · Electronics · Software · Control · Communications

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00 Overview

1. Purpose & Scope

This document provides a complete technical reference for the Inspection Robot system — an autonomous, mobile platform designed for non-destructive inspection of industrial infrastructure including pipelines, storage tanks, confined spaces, power-line towers, and structural steel in bridges or offshore platforms. The robot reduces human exposure to hazardous environments while delivering high-resolution sensor data for predictive maintenance workflows.

2. Robot Architecture

Subsystems	The robot integrates six primary subsystems: (1) Mechanical chassis and locomotion, (2) Sensor suite (cameras, LiDAR, ultrasonic, gas), (3) On-board processing and control, (4) Power and energy management, (5) Communication and telemetry, and (6) Software stack including ROS 2, SLAM and mission-planning modules.

3. Key Specifications

Parameter	Value	Notes
Overall Dimensions	450 × 320 × 180 mm	L × W × H (stowed)
Total Mass	8.5 kg	Including battery pack
Payload Capacity	1.2 kg	Additional sensor heads
Max Speed	1.2 m/s	On flat surface
Climbing Ability	45°	Magnetic track variant
Battery Life	~4 h	Typical inspection duty
IP Rating	IP65	Dust & water jet proof
Operating Temp.	-10 °C to +60 °C	Extended range optional
Comms Range	500 m LOS	Wi-Fi 5 / 900 MHz backup
CPU	NVIDIA Jetson Orin NX	16 GB unified memory
IMU Rate	1 kHz	9-DOF MEMS

01 Mechanical System

1.1 Frame & Chassis Design

The chassis is machined from 6061-T6 aluminium alloy with CNC-milled pockets to reduce weight while maintaining structural rigidity. Finite element analysis (FEA) was performed using Ansys Mechanical, achieving a safety factor of ≥ 3.5 under worst-case impact loads. Carbon-fibre composite panels form the outer enclosure, rated to IP65. M3/M4 stainless steel threaded inserts provide corrosion-resistant mounting points.

Component	Material	Mass (g)	Process
Main frame	6061-T6 Aluminium	1 820	CNC milling
Side panels	CF/Epoxy composite	310	Autoclave layup
Motor mounts	304 Stainless steel	240	Laser cut + bend
Top cover	Polycarbonate	185	CNC milling
Wheel hubs	POM Delrin	90	Turning

1.2 Locomotion Mechanism

- Differential-drive with four 100 mm polyurethane wheels. Front and rear axles are coupled via timing belt (3 mm GT2 pitch) to allow a single motor per side.
- An optional magnetic-track module (neodymium magnet array, pull force 120 N/module) enables vertical wall climbing on ferromagnetic surfaces.
- Passive compliance in each wheel hub absorbs surface irregularities up to 15 mm.
- Turning radius: 0 mm (point turn in place), maximum lateral payload shift < 5 mm.

1.3 Payload & Tooling Interface

A standardised NATO STANAG-style dovetail rail on the top deck accepts interchangeable sensor heads. Quick-release M5 bolts allow sensor swap in < 2 minutes. Electrical connection is via a 12-pin Amphenol circular connector (IP67) supplying 5 V/3 A, 12 V/5 A and two USB 3.2 data lines.

02 Motors

2.1 Drive Motors — Brushless DC (BLDC)

Each drive axle uses a T-Motor MN3508 KV700 BLDC motor. The high-pole-count design (14 poles) provides excellent low-speed torque, critical for slow inspection traversal. Hall-effect sensors embedded in the stator winding provide absolute commutation feedback.

Model	T-Motor MN3508 KV700
Stator	35 × 08 mm (diam. × height)
KV Rating	700 rpm/V
Peak Torque	0.32 N·m
Peak Current	16 A
Efficiency	≥ 89 % at rated load
Weight	98 g
Shaft Dia.	5 mm (D-flat)
Feedback	Hall-effect + back-EMF

2.2 Pan-Tilt Camera Servo Motors

The PTZ (pan-tilt-zoom) camera gimbal uses Dynamixel XM430-W350 smart servos. These digital actuators integrate a 4096-count absolute encoder, PID-adjustable torque control and real-time status feedback over RS-485 half-duplex UART.

Parameter	Specification
Model	Dynamixel XM430-W350-R
Stall Torque	4.1 N·m @ 12 V
No-load Speed	46 rpm
Position Resolution	0.088° (4096 counts/rev)
Communication	RS-485 half-duplex @ 4 Mbps
Operating Voltage	10.0 – 14.8 V
Protocol	Dynamixel Protocol 2.0

03 Motor Drivers

3.1 BLDC Field-Oriented Control (FOC) Driver

Drive motors are controlled by the ODrive S1 brushless motor controller, which implements a complete FOC pipeline: Clarke & Park transforms, space-vector PWM at 40 kHz, and closed-loop current/velocity/position modes. The controller connects to the main SBC via CAN FD at 5 Mbps.

Controller	ODrive S1
Input Voltage	12 – 56 V DC
Continuous I	40 A per axis
Peak I	120 A (1 s burst)
PWM Frequency	40 kHz
Control Loop	FOC — d-q axis decoupling
Interface	CAN FD / USB / UART
Feedback	Hall + encoder, ±0.01° positioning
Protections	OVP, OCP, OTP, short-circuit

3.2 H-Bridge Driver for Auxiliary Actuators

Auxiliary actuators (brush cleaning head, gripper) use the DRV8874 H-bridge IC from Texas Instruments. A 3.3 V-compatible PWM interface with integrated decay-mode control simplifies MCU integration.

Parameter	Value
Device	Texas Instruments DRV8874
Voltage Range	4.5 – 40 V
RDS(on)	130 mΩ (high-side + low-side)
Peak Current	8 A
PWM Freq. max	200 kHz
Logic Level	3.3 V / 5 V compatible
Package	HTSSOP-20 with thermal pad

04 Sensors

4.1 Vision — RGB-D Camera

An Intel RealSense D455 depth camera provides colour and structured-light depth imagery. It is mounted on the front face of the robot and used for SLAM mapping, obstacle avoidance and surface defect detection.

Model	Intel RealSense D455
RGB Resolution	1920 × 1080 @ 30 fps
Depth Resolution	1280 × 720 @ 30 fps
Depth Range	0.4 – 6 m (optimal 0.6 – 4 m)
Depth Accuracy	< 2 % at 4 m
IMU	BMI055 6-DOF @ 400 Hz
Interface	USB 3.2 Gen 1
Power	< 1.5 W nominal

4.2 LiDAR — 2D Scanning

A RPLIDAR A3 spinning LiDAR handles 2D planar mapping at 16 000 samples/s. It is used alongside the D455 for robust localisation in GPS-denied environments.

Model	Slamtec RPLIDAR A3
Range	0.2 – 25 m
Scan Rate	10 Hz (5 Hz optional)
Angular Res.	0.225° (@ 10 Hz)
Samples/scan	16 000
Laser Class	Class 1 (eye-safe)
Interface	UART @ 256 kbps
Power	5 V / 0.5 A

4.3 Environmental & Gas Sensors

Sensor	Parameter	Range	Output
Sensirion STS40	Temperature	-40 to +125 °C	I²C
Sensirion SHT45	Humidity	0 – 100 % RH	I²C
MiCS-6814	CO / NO2 / NH3	1 – 1000 ppm	ADC (0-3 V)
SGP40	VOC Index	0 – 500 (index)	I²C
Bosch BMP388	Pressure/Altitude	300 – 1100 hPa	SPI/I²C

4.4 Ultrasonic Proximity Array

Six HC-SR04 ultrasonic sensors form a 360° collision-avoidance ring at 30° intervals. Each sensor is polled by a dedicated STM32G0 co-processor to avoid GPIO timing constraints on the main CPU. Range: 2 cm – 4 m, accuracy ±3 mm.

05 Control System

5.1 Embedded Controller Architecture

The main compute platform is the NVIDIA Jetson Orin NX (16 GB). A secondary STM32H743 MCU runs a hard real-time control loop (1 kHz) and acts as a safety watchdog, capable of halting all motors independently of the main SBC.

Module	Role	OS / Firmware	Update Rate
Jetson Orin NX	Mission & perception	Ubuntu 22.04 + ROS 2 Humble	30 Hz vision
STM32H743	Hard real-time control	FreeRTOS 10.x	1 kHz motor
STM32G0 (×6)	Ultrasonic polling	Bare-metal	50 Hz each
ODrive S1 (×2)	FOC motor drive	ODrive firmware 0.6.x	40 kHz PWM

5.2 Sensor Fusion

An Extended Kalman Filter (EKF) fuses data from the IMU (1 kHz), wheel odometry (500 Hz) and LiDAR scan-matching (10 Hz) to produce a 6-DOF pose estimate with covariance. The robot_localization ROS 2 package (navsat_transform disabled) implements this pipeline. Expected position drift < 0.5 % of distance travelled.

06 Power System

6.1 Battery Technology

The primary energy source is a custom 6S4P lithium-ion pack using Samsung 50S 21700 cells (5 000 mAh each). The pack delivers 22.2 V nominal, 20 Ah capacity (444 Wh), sufficient for approximately 4 hours at typical inspection duty. A dedicated battery management system (BMS) monitors cell voltages, temperatures, and state-of-charge with ±1 % SoC accuracy.

Chemistry	Li-ion (INR21700-50S)
Configuration	6S4P — 22.2 V nominal, 25.2 V full
Capacity	20 Ah / 444 Wh
Max Discharge	60 A continuous (3C)
BMS Balancer	Passive, ±5 mV balance threshold
Charge Method	CC-CV, 4 A (0.2C) standard
Cycle Life	≥ 500 cycles to 80 % capacity
Weight	1.8 kg

6.2 Power Distribution

Rail	Voltage	Max Current	Load
VBAT (raw)	22.2 V (nom.)	60 A	Motor drivers
V12	12.0 V	10 A	Servos, LiDAR, fans
V5	5.0 V	8 A	SBC I/O, cameras, USB hubs
V3V3	3.3 V	3 A	MCU, sensors, logic
VBAT_PAYLOAD	22.2 V (fused)	5 A	Payload connector

07 Communication

7.1 Wireless Link — Primary

Primary telemetry and control uses a dual-band Wi-Fi 6 (802.11ax) radio module (Intel AX210) paired with a 5 dBi directional patch antenna. A dedicated GCS (Ground Control Station) laptop running custom Qt5 HMI connects via a private 802.11ax access point at 5 GHz. Typical throughput: 150 Mbps (video + telemetry).

7.2 Fallback — 900 MHz FHSS

A Digi XBee SX 900 MHz frequency-hopping spread-spectrum radio provides a redundant control uplink at up to 500 m LOS in cluttered industrial environments where 5 GHz signals attenuate severely. Bandwidth is limited to MAVLink telemetry (57.6 kbps) on this channel.

7.3 Wired & On-Board Protocols

Protocol	Bus	Speed	Use
CAN FD	2-wire diff.	5 Mbps	Motor controllers ↔ STM32
RS-485	2-wire diff.	4 Mbps	Dynamixel servo chain
I²C	2-wire	400 kHz	Environmental sensors
SPI	4-wire	8 MHz	BMP388 pressure, flash NOR
UART	2-wire	256 kbps	RPLIDAR, GPS (optional)
USB 3.2	Differential	5 Gbps	RealSense D455, debug
Ethernet	RJ45 (int.)	1 GbE	Jetson ↔ STM32 Ethernet

08 Kinematics

8.1 Differential Drive Kinematics

The robot uses a differential drive model. Let v_L and v_R be the left and right wheel tangential velocities, b the track width (0.28 m), and r the wheel radius (0.05 m). The linear velocity v and angular velocity w of the robot centre are:

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v = (v_R + v_L) / 2     w = (v_R - v_L) / b
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The forward kinematic model integrates pose (x, y, theta) from odometry at 500 Hz, producing dead-reckoning estimates corrected by the EKF at 10 Hz.

8.2 PTZ Camera Gimbal — 2-DOF Kinematics

Pan (rotation about world Z-axis) and tilt (rotation about body Y-axis) are modelled with Denavit-Hartenberg parameters. The workspace envelope is: Pan ±180°, Tilt -30° to +90°. Inverse kinematics for a target 3D point P_world are solved analytically via atan2 without singularity in the reachable workspace.

Joint	DOF	Range	Max Speed	Torque
Pan	Rotation Z	±180°	60°/s	4.1 N·m
Tilt	Rotation Y	-30° to +90°	60°/s	4.1 N·m

09 Control Algorithms

9.1 Velocity PID Controller

Each wheel uses an independent PID velocity controller running at 1 kHz on the STM32H743. The control law is: $u(t) = Kp \cdot e(t) + Ki \cdot \int e\, dt + Kd \cdot de/dt$, where $e = v_{ref} - v_{measured}$. Anti-windup is implemented via integrator clamping. Tuned gains: $Kp = 2.1$, $Ki = 0.85$, $Kd = 0.04$.

9.2 Path Following — Pure Pursuit

High-level path following uses the Pure Pursuit algorithm. A look-ahead distance L_d (adaptive, 0.3–0.8 m based on speed) selects a waypoint on the planned path. The required curvature $kappa = 2 \cdot \sin(alpha) / L_d$, where alpha is the heading error to the look-ahead point, is converted to differential-drive wheel speeds.

9.3 SLAM — Cartographer 2D

Google Cartographer (2D mode) processes RPLIDAR scans and fused odometry to build an occupancy grid map. Loop-closure is detected via branch-and-bound scan matching with a 0.1 m^2 acceptance threshold. Maps are saved as YAML + PGM for re-use.

11 Electronics

11.1 Main PCB — Robot Brain Board

A custom 4-layer PCB (100 × 80 mm, FR4, 1 oz copper) integrates the STM32H743 MCU, CAN FD transceivers (TCAN1042), RS-485 driver (SN65HVD72), I²C multiplexer (TCA9548A), 3.3 V/5 V LDO regulators, and a 16-pin FFC connector for the sensor breakout board. Designed in KiCad 7; Gerber files are in the /11_Electronics/ folder.

11.2 Signal Conditioning & EMI

- All CAN FD and RS-485 differential pairs are impedance-controlled (120 Ω ±10 %).
- A ground plane splits power (PWR_GND) and signal (SIG_GND) domains, joined at one star point near the main connector to prevent ground-loop interference.
- Ferrite beads (600 Ω @ 100 MHz) are placed on the VBAT lines entering each PCB.
- TVS diodes (SMBJ12A) on all external connector pins provide transient protection.
- Chassis is bonded to SIG_GND via a 1 Ω / 2 W chassis-ground resistor.

12 Software

12.1 Software Architecture

The software stack is built on ROS 2 Humble (Ubuntu 22.04). The node graph follows a layered architecture: hardware abstraction nodes → perception nodes → navigation nodes → mission nodes. All inter-node communication uses DDS (CycloneDX) with QoS policies tuned per data type (reliable for control, best-effort for high-rate sensor streams).

12.2 Key ROS 2 Packages

Package	Function	Language
robot_bringup	Launch files & parameter management	Python
hw_interface	STM32 CAN FD bridge node	C++
sensor_driver	D455, RPLIDAR, gas sensor nodes	C++/Python
robot_localization	EKF pose fusion	C++
slam_toolbox	Lifelong SLAM mapping	C++
nav2	Path planning & following	C++
defect_detector	YOLOv8 corrosion/crack detection	Python
mission_manager	BT-based mission sequencer	C++
gcs_bridge	WebSocket telemetry to GCS	Python

12.3 Computer Vision Pipeline

Defect detection uses a YOLOv8-s model fine-tuned on a proprietary dataset of 3 200 annotated images of corrosion, cracks, weld spatter, and delamination. Inference runs on the Jetson Orin NX GPU at 28 fps (1280×720). Detected bounding boxes are geo-referenced using the D455 depth map and EKF pose, creating a 3D defect map exported as a GeoJSON file per mission.

13 Manufacturing

13.1 Materials & Tolerances

Part Class	Material	Key Tolerance	Surface Finish
Structural brackets	6061-T6 Al	±0.1 mm	Anodised type II
Motor mounts	304 SS	±0.05 mm	Electropolished
Wheel hubs	POM Delrin	±0.02 mm	Turned, Ra 1.6
Enclosure panels	CF/Epoxy	±0.5 mm	Clear lacquer
Fasteners	A2 SS M3/M4/M5	ISO 4762	DIN 934 hex

13.2 Assembly Sequence

Step 1 — Frame Assembly	Press all threaded inserts into the main Al frame. Verify flatness < 0.3 mm.
Step 2 — Drivetrain	Mount motor housings; install timing belts with 10 N pre-tension. Check backlash.
Step 3 — PCB Installation	Torque M3 PCB standoffs to 0.4 N·m. Connect harnesses before closing.
Step 4 — Sensor Mounting	Align D455 to centreline ±1°. Secure RPLIDAR with vibration isolators.
Step 5 — Battery Fit	Slide battery pack; secure latch; connect XT60 and BMS balance leads.
Step 6 — Enclosure Seal	Apply Loctite 5188 RTV to all panel edges. Torque cover screws M–O pattern.
Step 7 — Functional Test	Run bench test sequence (Section 12) before field deployment.

14 Research Papers

Key Reference Literature

SLAM & Navigation

Hess et al. (2016) 'Real-Time Loop Closure in 2D LIDAR SLAM,' ICRA 2016. — Foundational paper for Cartographer algorithm used in this system.

Visual Odometry

Engel et al. (2018) 'Direct Sparse Odometry,' IEEE TPAMI. — Basis for dense depth tracking in camera-only fallback mode.

Defect Detection

Deng et al. (2022) 'Corrosion Detection in Industrial Pipelines Using Deep Convolutional Neural Networks,' NDT&E; International. — Training methodology adapted for the YOLOv8 defect model.

Magnetic Climbing

Zhang et al. (2021) 'Design and Analysis of Permanent Magnet Wheeled Climbing Robot for Ship Hull Inspection,' Robotics and Autonomous Systems.

EKF Sensor Fusion

Moore & Stouch (2014) 'A Generalized Extended Kalman Filter Implementation for the Robot Operating System,' IIAS. — robot_localization package basis.

FOC Motor Control

Holtz (1994) 'Pulsewidth Modulation for Electronic Power Conversion,' Proceedings IEEE. — Theoretical basis for the ODrive FOC implementation.

15 Datasheets

Component Index with Critical Ratings

Part No.	Component	Mfr.	Vceo/Vmax	Imax	Package
MN3508	BLDC Motor	T-Motor	—	16 A	Outrunner
XM430-W350	Smart Servo	Robotis	14.8 V	2.1 A	Custom
ODrive S1	FOC Controller	ODrive	56 V	40 A	PCB module
DRV8874	H-Bridge Driver	TI	40 V	8 A	HTSSOP-20
STM32H743VIT6	MCU 480 MHz	ST	3.6 V	—	LQFP-100
Jetson Orin NX	SBC + GPU	NVIDIA	5 V	25 W	Module
D455	RGB-D Camera	Intel	5 V	1.5 A	USB-C
RPLIDAR A3	2D LiDAR	Slamtec	5 V	0.5 A	Custom
AX210	Wi-Fi 6 / BT 5.2	Intel	3.3 V	—	M.2 2230
TCAN1042	CAN FD Transceiver	TI	5.5 V	—	SOIC-8
SN65HVD72	RS-485 Transceiver	TI	5.5 V	—	SOIC-8
Samsung 50S	Li-ion Cell 21700	Samsung SDI	4.25 V	15 A	21700
MiCS-6814	Multi-Gas Sensor	SGX	5 V	—	SMD
SGP40	VOC Sensor	Sensirion	3.3 V	—	DFN-6
BMP388	Barometric Pressure	Bosch	3.6 V	—	LGA-8

16 Metadata

components.csv — Schema

The components.csv file contains one row per BOM item with the following columns:

Column	Type	Description
component_id	STRING PK	Unique part identifier e.g. C-MN3508-001
name	STRING	Human-readable component name
manufacturer	STRING	OEM manufacturer
part_number	STRING	Manufacturer part number
quantity	INT	Quantity per robot build
supplier	STRING	Preferred procurement source
unit_cost_usd	FLOAT	Last known unit cost in USD
subsystem	STRING	Section code e.g. 02_Motors
lead_time_days	INT	Typical delivery lead time
notes	STRING	Substitutes, hazards, storage notes

documents.csv — Schema

Column	Type	Description
doc_id	STRING PK	Document identifier e.g. D-12-SW-001
title	STRING	Document title
section	STRING	Associated section code
file_path	STRING	Relative path within repository
version	STRING	Semantic version e.g. 1.2.0
author	STRING	Primary author
date_updated	DATE	ISO 8601 last-modified date
status	ENUM	Draft / Review / Released / Obsolete

sources.csv — Schema

Column	Type	Description
source_id	STRING PK	Citation key e.g. REF-SLAM-001

type	ENUM	Paper / Datasheet / Standard / Book / URL
title	STRING	Full citation title
authors	STRING	Comma-separated author list
year	INT	Publication year
doi_url	STRING	DOI or URL if available
section_ref	STRING	Section where cited